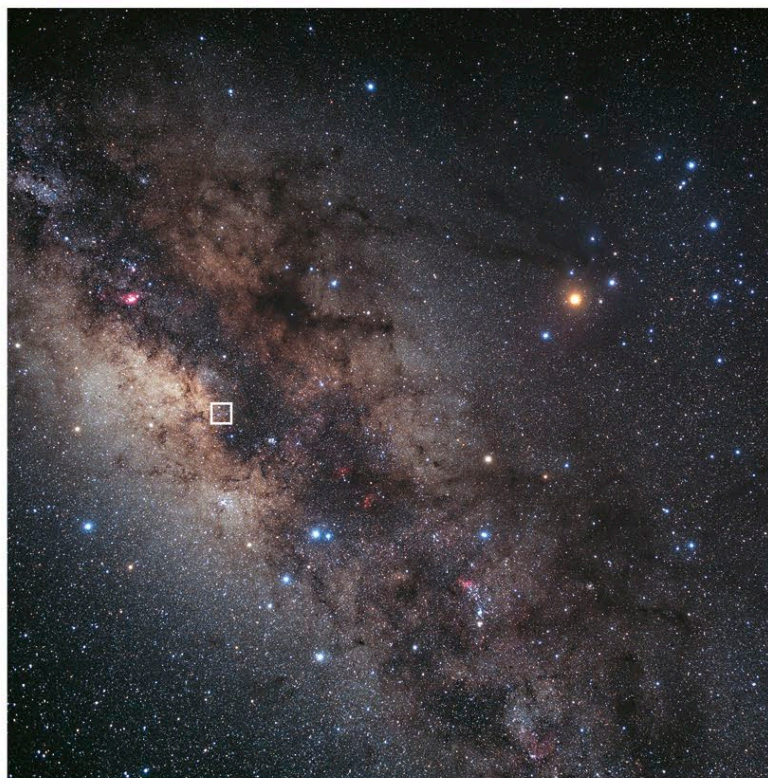




TOWARD THE CENTER OF THE GALAXY

The center of the galaxy cannot be seen directly, because it is obscured behind the dense Milky Way star fields of Sagittarius and Scorpius. The exact center is thought to be marked by an intense radio source called Sagittarius A* (boxed).

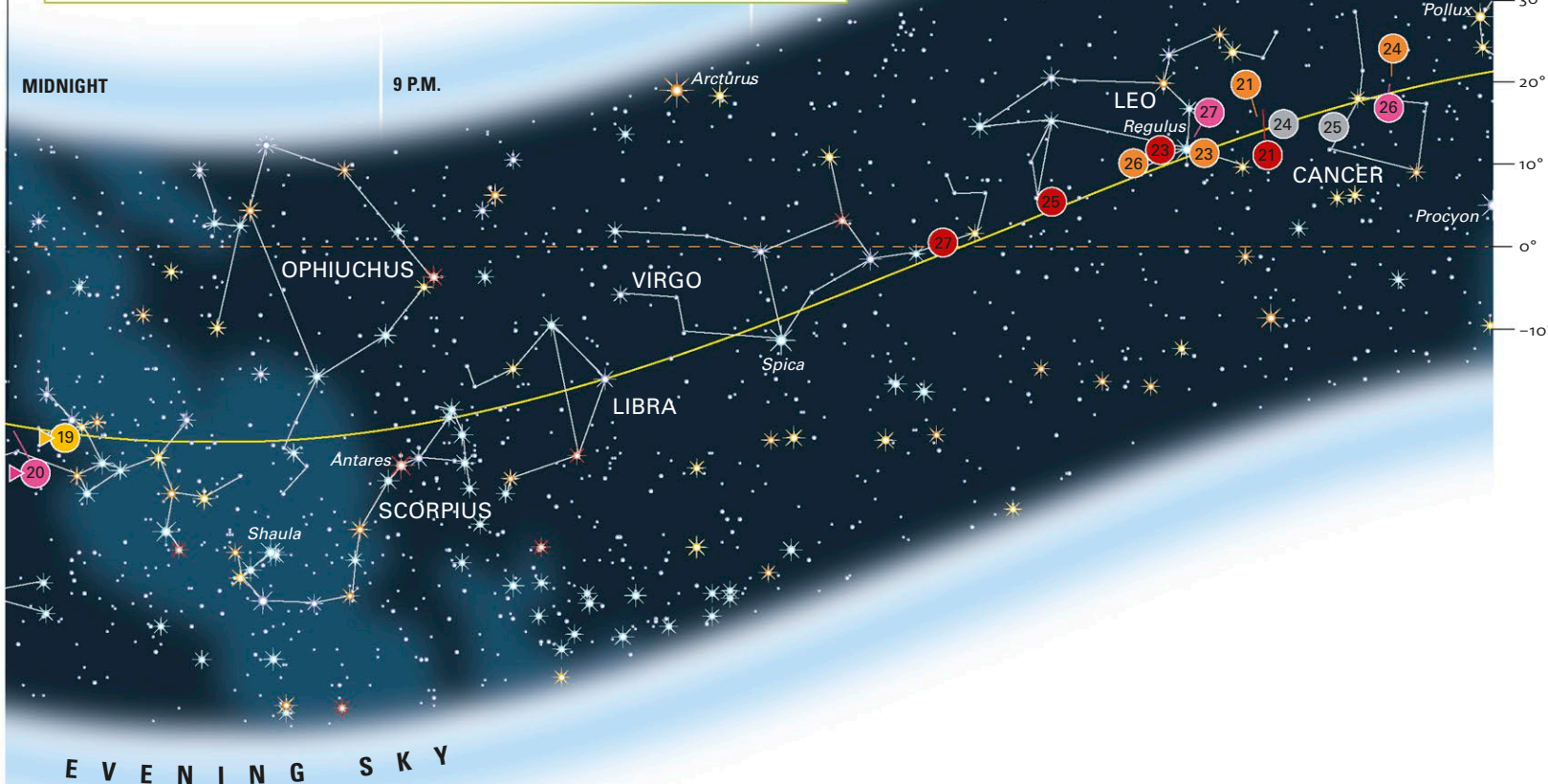
POSITIONS OF THE PLANETS

This chart shows the positions of the planets in July from 2020 to 2027. The planets are represented by colored dots, while the number inside each dot denotes the year. For all planets apart from Mercury, the dot indicates the planet's position on July 15. Mercury is shown only when it is at greatest elongation (see p.68)—for the specific date, refer to the table, left.

Mercury	Mars	Saturn	Neptune
Venus	Jupiter	Uranus	

EXAMPLES

- 25 Jupiter's position on July 15, 2025
- 21 Jupiter's position on July 15, 2021. The arrow indicates that the planet is in retrograde motion (see p.68).



SOUTHERN LATITUDES

THE STARS

The curved tail of Scorpius and the asterism known as the Teapot, formed from the main stars of Sagittarius, are virtually overhead for southern observers. The Milky Way is particularly dense and bright toward Sagittarius and Scorpius because this is the view toward the center of the galaxy.

Alpha (α) and Beta (β) Centauri—Rigel Kentaurus (see p.252) and Hadar—are in the southwest, pointing down to Crux, the Southern Cross. Spica (in Virgo) is in the eastern sky; Arcturus (in Boötes) is in the northwest; and Vega (see p.253), in Lyra, is in the north. Altair (see p.252), in Aquila, is high in the northeast, and observers about 30°S or closer to the equator can see Deneb, in Cygnus, low in the northeast. In the southeast, 1st-magnitude Fomalhaut, in Piscis Austrinus, enters the scene.

DEEP-SKY OBJECTS

Sagittarius is well stocked with outstanding deep-sky objects, among them the 5th-magnitude globular cluster M22, which is visible to the naked eye

under good conditions. The Lagoon Nebula (see p.243), or M8, an elongated gas cloud containing the star cluster NGC 6530, can be seen well through binoculars. To the north, in Serpens Cauda, the tail of the Serpent, lies the cluster M16—visible through binoculars—embedded in the much fainter Eagle Nebula (see pp.244–245).

Other famous deep-sky objects in Sagittarius, such as the Trifid Nebula, M20 (see p.246), need to be seen through a telescope. However, one particularly bright patch of the Milky Way, M24, is prominent to the naked eye. In adjoining Scorpius, the bright open clusters M6 and M7 remain high in the sky.

METEOR SHOWER

The Delta Aquariids, the best southern meteor shower, is active in July and August, reaching a peak around July 29. At best, perhaps 20 meteors an hour can be seen radiating from the southern half of Aquarius, but they are not particularly bright.